Genuine Multipartite Entanglement Analysis

Hubbard ring - Convex-roof framework

System parameters

Model	**Hubbard 1D-PBC**
Mode	**uniform**
Sites (L)	**4**
Electrons (Ne)	**4**
Hopping (t)	**1.0**
Interaction (U)	**4.0**
n-body order	**4**
Eigenstate idx	**0 (ground state)**

Eigenstate properties

Energy E0	**-2.10274848**
Energy E1	**-1.80642385**
Gap (E1-E0)	**0.296325**
<S^2>	**0.000000**
S	**0.0000**
<Sz>	**0.000000**
Multiplicity	**1.0**
Spin label	**Singlet (S=0)**

Double occupancy

<D> total	**0.287325**
<D> / site	**0.071831**

RDM and GME statistics for this run (4-body clusters)

Hilbert space dim : 2^4 = 16
Number of clusters : 4
Pure clusters : 4
Mixed clusters : 0
Clusters with GME (>0) : 4

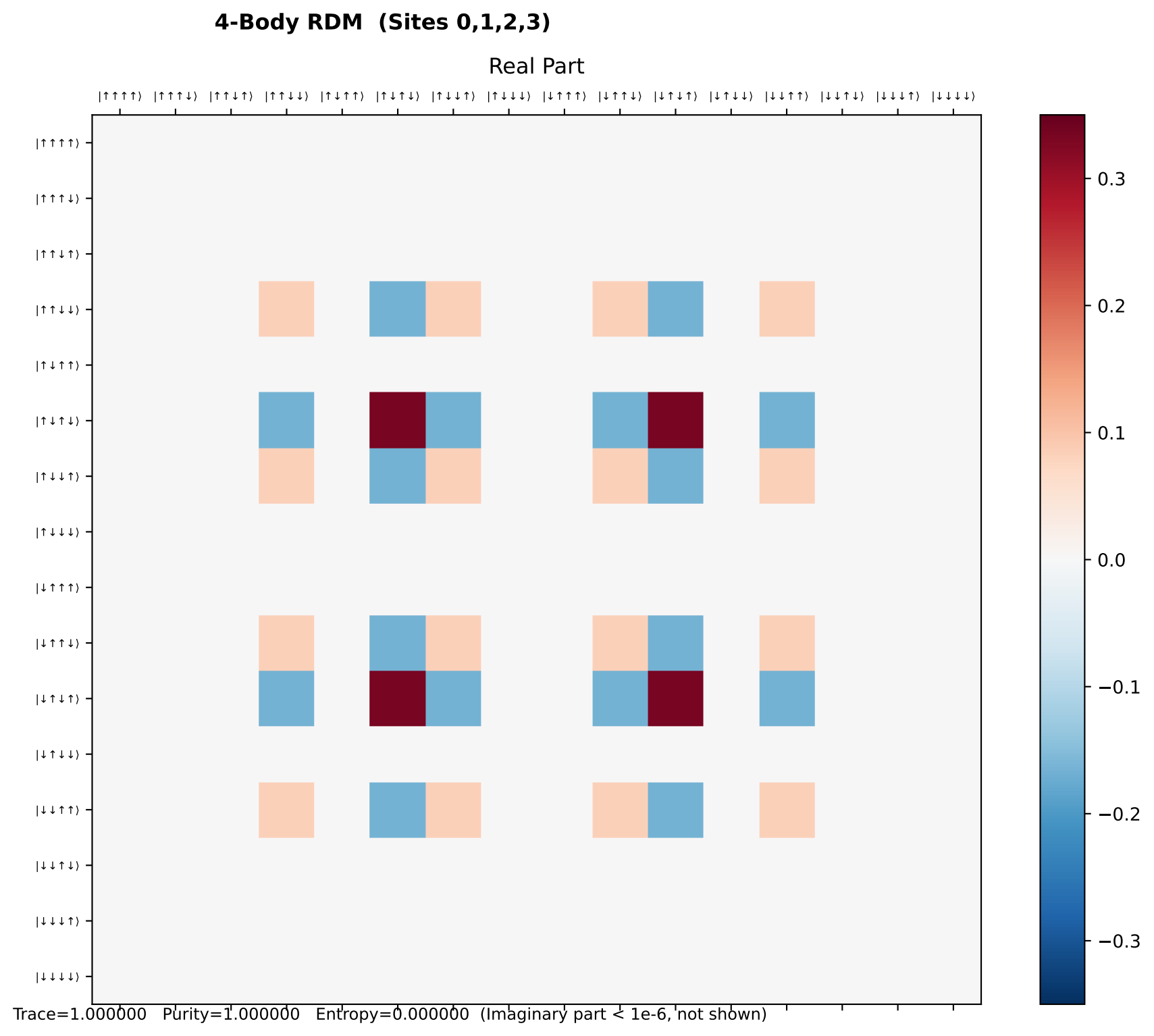
Purity gamma = Tr(rho^2) : mean = 1.000000 [min 1.000000 , max 1.000000]
Von Neumann entropy : mean = 0.000000 [min 0.000000 , max 0.000000]
Rank of rho : mean = 1.00 [min 1 , max 1]
C_GME (best estimate) : mean = 0.912871 [min 0.912871 , max 0.912871]

Summary: All 4-Body RDM Clusters

System: L=4, Ne=4, U=4.0, t=1.0 | E_0=-2.102748

Sites	Purity	C_GME	C_LB	C_cayl	C_expo	C_stiefel	C_UB	Status
(0, 1, 2, 3)	1.000000	0.912871	0.000000	0.000000	0.000000	0.912871	0.912871	GME
(1, 2, 3, 0)	1.000000	0.912871	0.000000	0.000000	0.000000	0.912871	0.912871	GME
(2, 3, 0, 1)	1.000000	0.912871	0.000000	0.000000	0.000000	0.912871	0.912871	GME
(3, 0, 1, 2)	1.000000	0.912871	0.000000	0.000000	0.000000	0.912871	0.912871	GME

Legend:
  Purity = Tr(rho²) [1.0=pure, <1.0=mixed] | C_GME = Genuine multipartite entanglement (pure only)
  C_LB = Lower bound (notes eqs.6-9) | C_cayl = unitary optimization - Cayley Transform |C_expo= unitary optimization - Expoenetial Map | BEST
  C_UB = Upper bound (eigendecomp) | Status: GME =has GME | NO GME=separable | MIXED=mixed state


4-Body RDM (Sites 0,1,2,3)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 0,1,2,3)

```
RDM Metrics
==================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 1.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -4.749088e-17
  Eigenvalue max       : 1.000000e+00
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
==================================

  gamma = Tr(rho^2) = 1.00000000

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------
  (0,)         | (1, 2, 3)    | 0.50000000
  (0, 1)       | (2, 3)       | 0.58333333
  (0, 2)       | (1, 3)       | 0.33333333
  (0, 3)       | (1, 2)       | 0.58333333
  (0, 1, 2)    | (3,)         | 0.50000000
  (0, 1, 3)    | (2,)         | 0.50000000
  (0, 2, 3)    | (1,)         | 0.50000000

STATUS: GME EXISTS
----------------------------------

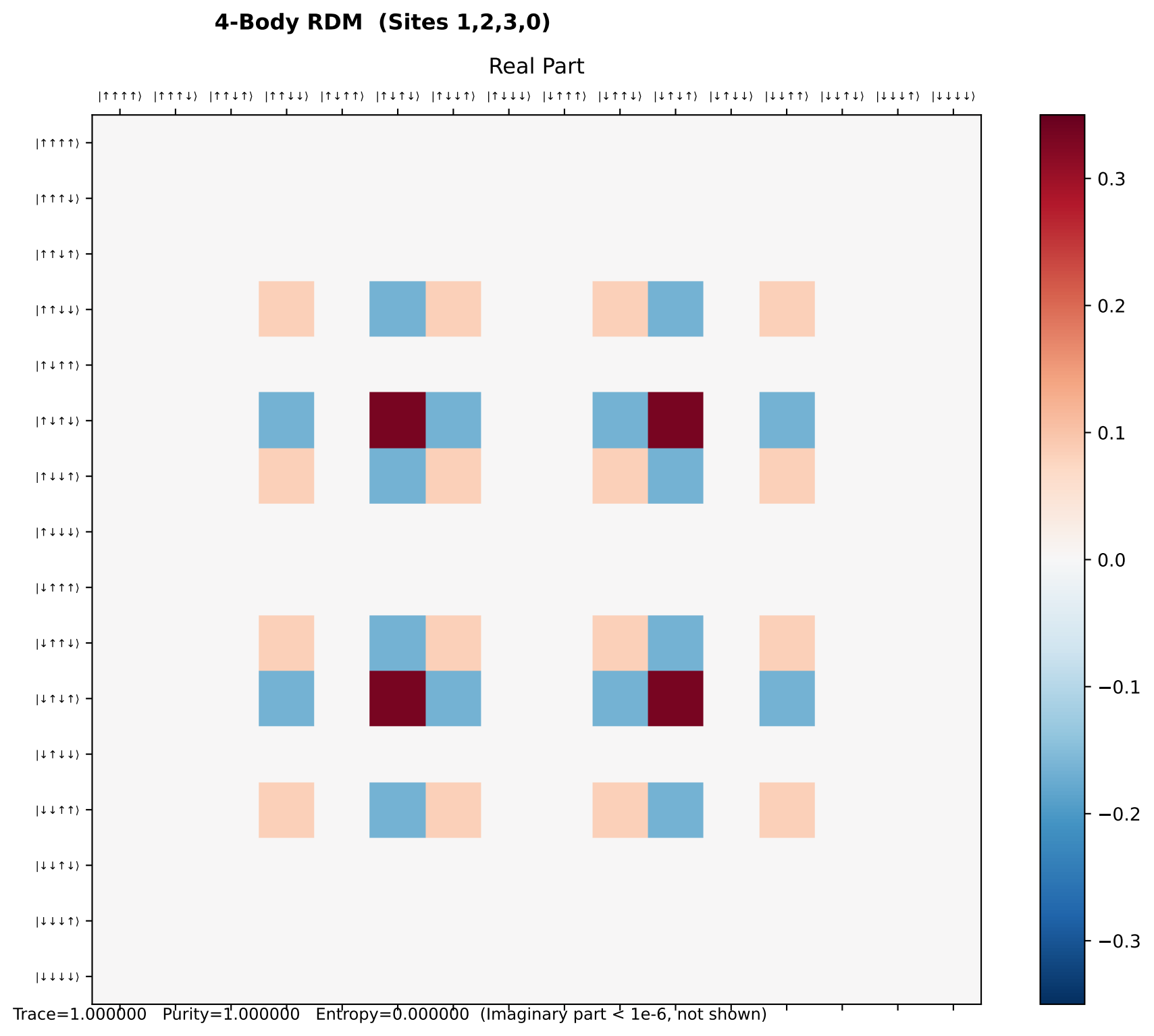
C_GME = 0.91287093
GME confirmed across ALL cuts.
lower bound [eqs.6-9]
  --------------------------------

  C_exact = 0.91287093
  C_LB    = 0.66666667
  Gap     = 0.24620426
```

```
Convex Roof Analysis
==================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.91287093   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.91287093

  Status: GME
```


4-Body RDM (Sites 1,2,3,0)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 1,2,3,0)

```
RDM Metrics
==================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 1.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -7.045364e-17
  Eigenvalue max       : 1.000000e+00
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 1.00000000

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------
  (1,)         | (2, 3, 0)    | 0.50000000
  (1, 2)       | (3, 0)       | 0.58333333
  (1, 3)       | (2, 0)       | 0.33333333
  (1, 0)       | (2, 3)       | 0.58333333
  (1, 2, 3)    | (0,)         | 0.50000000
  (1, 2, 0)    | (3,)         | 0.50000000
  (1, 3, 0)    | (2,)         | 0.50000000

STATUS: GME EXISTS
------------------------------------

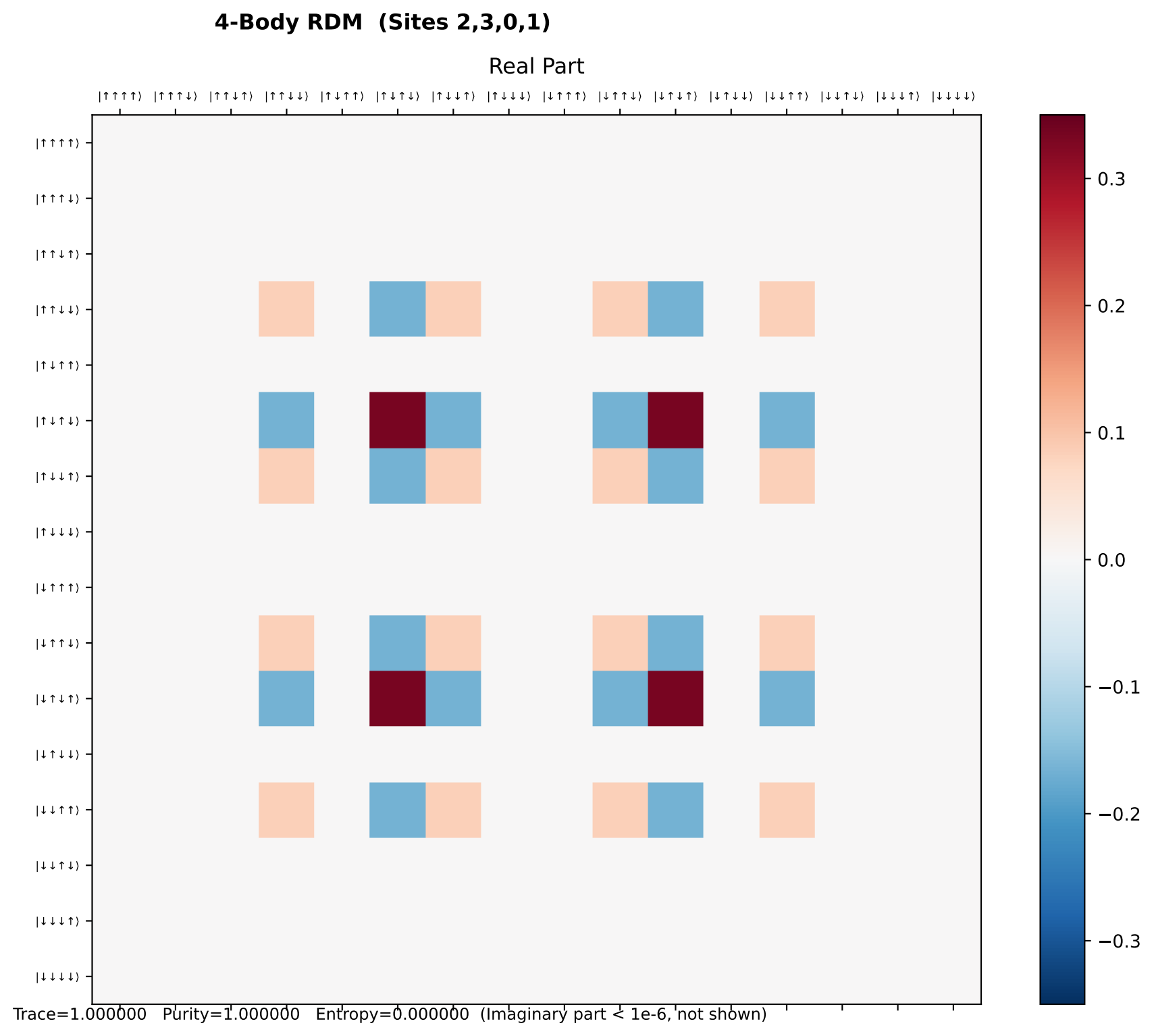
C_GME = 0.91287093
GME confirmed across ALL cuts.
lower bound [eqs.6-9]
  ----------------------------------

  C_exact = 0.91287093
  C_LB    = 0.66666667
  Gap     = 0.24620426
```

```
Convex Roof Analysis
==================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.91287093   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.91287093

  Status: GME
```


4-Body RDM (Sites 2,3,0,1)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 2,3,0,1)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 1.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -1.068409e-16
  Eigenvalue max       : 1.000000e+00
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 1.00000000

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------
  (2,)         | (3, 0, 1)    | 0.50000000
  (2, 3)       | (0, 1)       | 0.58333333
  (2, 0)       | (3, 1)       | 0.33333333
  (2, 1)       | (3, 0)       | 0.58333333
  (2, 3, 0)    | (1,)         | 0.50000000
  (2, 3, 1)    | (0,)         | 0.50000000
  (2, 0, 1)    | (3,)         | 0.50000000

STATUS: GME EXISTS
------------------------------------

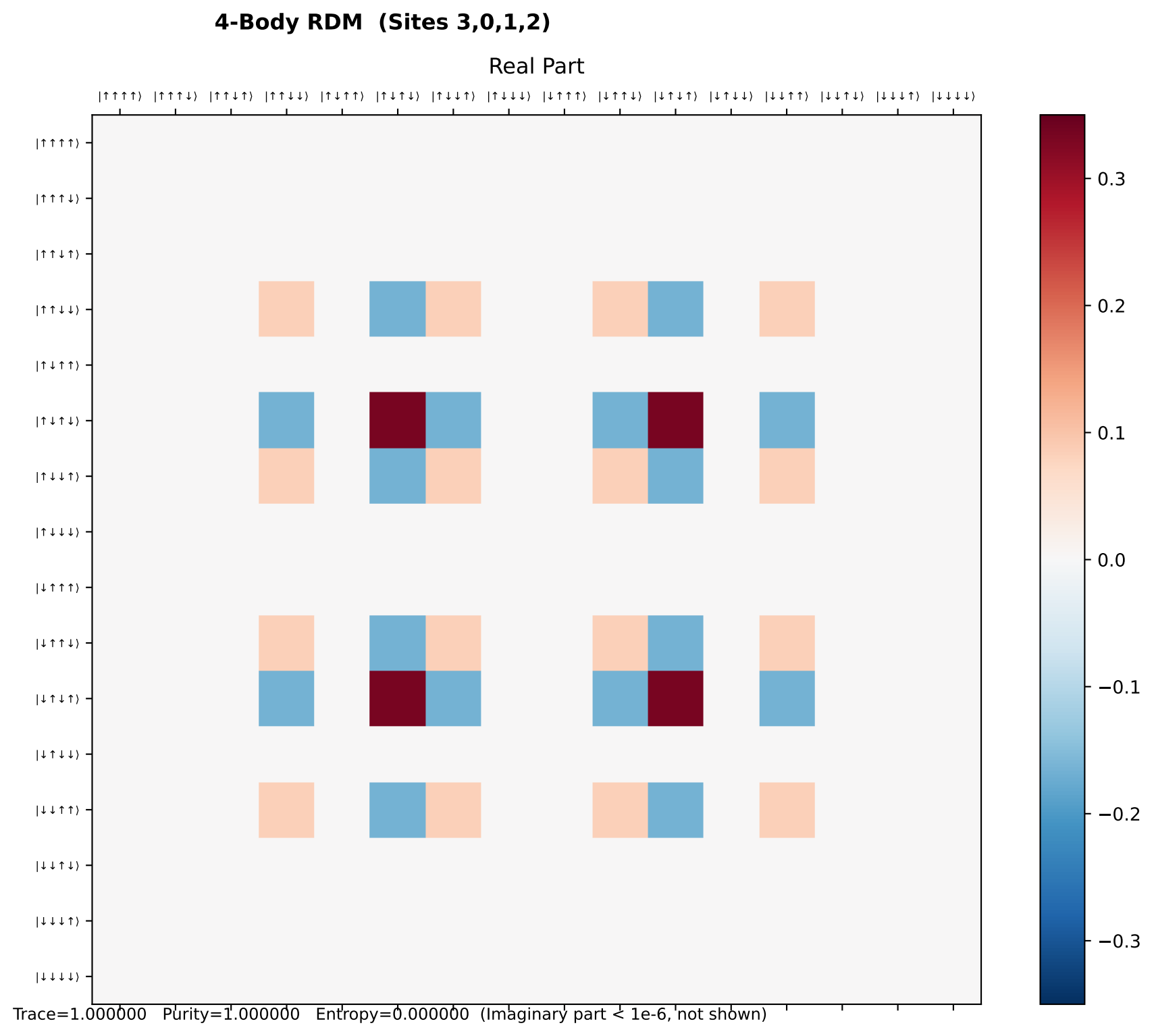
C_GME = 0.91287093
GME confirmed across ALL cuts.
lower bound [eqs.6-9]
  ----------------------------------

  C_exact = 0.91287093
  C_LB    = 0.66666667
  Gap     = 0.24620426
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.91287093   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.91287093

  Status: GME
```


4-Body RDM (Sites 3,0,1,2)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 3,0,1,2)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 1.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -9.909913e-17
  Eigenvalue max       : 1.000000e+00
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 1.00000000

  A            | B            | Tr(rho_A^2)
  -------------|--------------|------------
  (3,)         | (0, 1, 2)    | 0.50000000
  (3, 0)       | (1, 2)       | 0.58333333
  (3, 1)       | (0, 2)       | 0.33333333
  (3, 2)       | (0, 1)       | 0.58333333
  (3, 0, 1)    | (2,)         | 0.50000000
  (3, 0, 2)    | (1,)         | 0.50000000
  (3, 1, 2)    | (0,)         | 0.50000000

STATUS: GME EXISTS
------------------------------------

C_GME = 0.91287093
GME confirmed across ALL cuts.
lower bound [eqs.6-9]
  ----------------------------------

  C_exact = 0.91287093
  C_LB    = 0.66666667
  Gap     = 0.24620426
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.91287093   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.91287093

  Status: GME
```